

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 13 & Lecture 14

(Ch. 9)

The *IS-LM-PC* model

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model
 1. Medium and short run equilibrium
 2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

INTRODUCTION

- **Short Run**
 - Goods market and financial markets in equilibrium (IS-LM)
 - Expectations are not correct
 - There is a O.G.: actual output $\neq$ potential output
- **Medium Run**
 - Labor market is in equilibrium (WS=PS)
 - Expectations are correct
 - There is no O.G.: actual output = potential output
- **From short to medium run:**
 - Adjustment of expectations, prices and wages (PC)

IS-LM-PC MODEL

- Simply put together our short run and medium run models
 1. describe the transition from short to medium run
 2. identify the level of three endogenous variables for the economy
 1. output (Y)
 2. interest rate (i and r)
 3. inflation (π)
 3. understand if shocks produce temporary or permanent effects

IS extended: $Y = C(Y - T) + I(Y, r + x) + G$  the relevant interest rate for demand is the borrowing rate. r can be negative

LM: $r = \bar{r}$  CB sets i and indirectly controls r , given expectations

PC: $\pi - \pi^e = -\alpha(u - u_n)$  In the IS-LM model we explain $Y \Rightarrow$ necessary to rewrite the PC in terms of production, Y

PC IN TERMS OF OUTPUT GAP

Recalling that

$$Y_n = L(1 - u_n) \Rightarrow u_n = 1 - \frac{Y_n}{L}$$

$$u_t - u_n = 1 - \frac{Y_t}{L} - 1 + \frac{Y_n}{L} = -\frac{1}{L}(Y_t - Y_n)$$

The general Phillips curve becomes:

$$\pi - \pi^e = -\alpha(u_t - u_n)$$

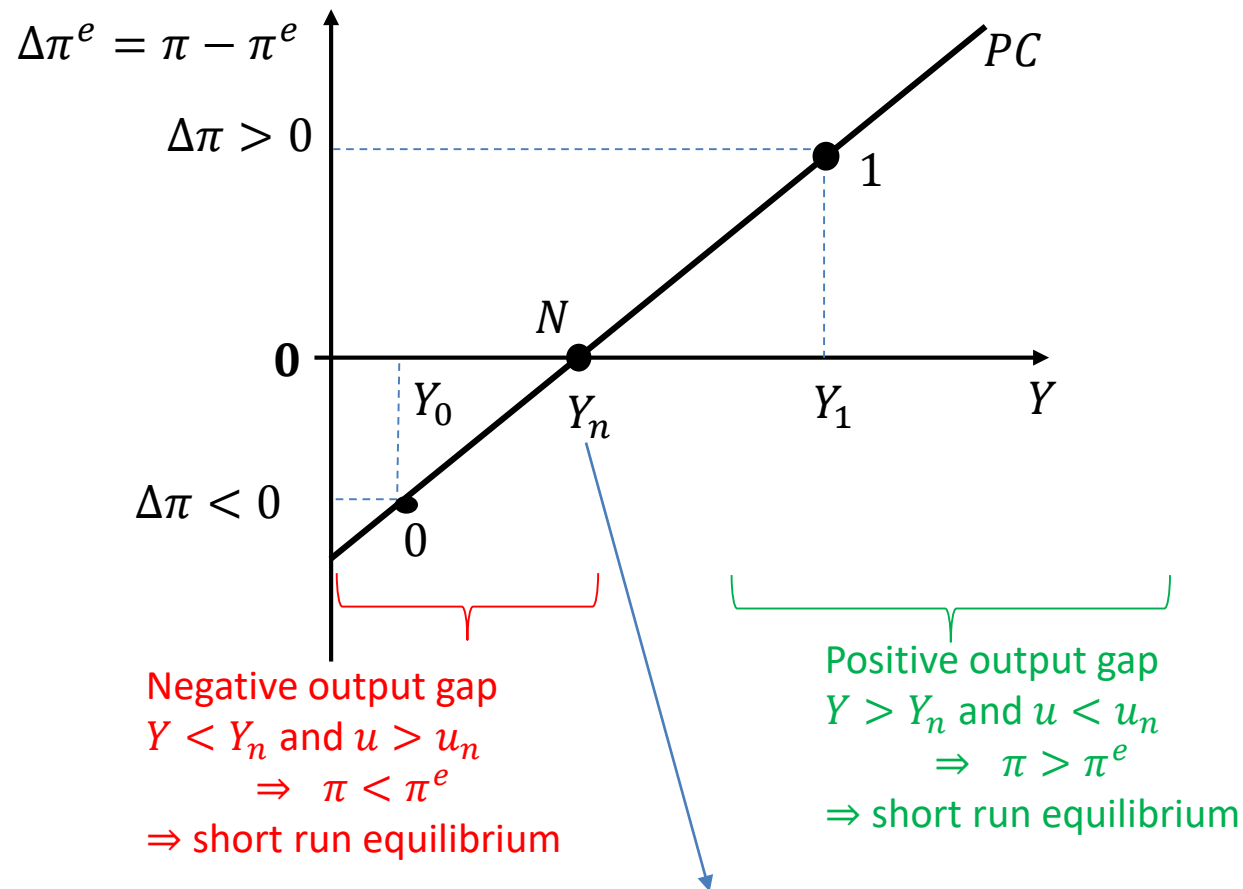
$$\Delta\pi^e = \pi - \pi^e = \frac{\alpha}{L}(Y_t - Y_n)$$

PC in terms of output gap

Describes the dynamics of prices as production changes wrt the potential

Agents can follow different rules to form expectations

Novelty: plot the PC CURVE in the $(Y, \Delta\pi^e)$ plane $\Delta\pi^e = \pi - \pi^e = \frac{\alpha}{L} (Y - Y_n)$
 +



The economy at the medium run
 equilibrium $Y = Y_n$ and $u = u_n$
 $\Rightarrow \pi = \pi^e$

NOTE on EXPECTATIONS

- General case: expectations anchored to the CB inflation target

$$\pi^e = \bar{\pi}$$

$$\Delta\pi^e = \pi - \bar{\pi} = \frac{\alpha}{L} (Y - Y_n)$$

With positive O.G. current inflation is above the CB target but it is constant

With negative O.G. current inflation is below the CB target but it is constant

With no O.G. current inflation is at the CB target

- Alternative case: adaptive expectations

$$\pi^e = \pi_{t-1}$$

$$\Delta\pi^e = \pi - \pi_{t-1} = \frac{\alpha}{L} (Y - Y_n)$$

With positive O.G. inflation increases over time

With negative O.G. inflation decreases over time

With no O.G. inflation is constant over time

Model endogenous variables: $Y, i(\text{and } r), \pi$

MEDIUM RUN EQUILIBRIUM

All markets (goods, financial, labor) are in equilibrium and expectations are correct => the economy is at its potential with no output gap

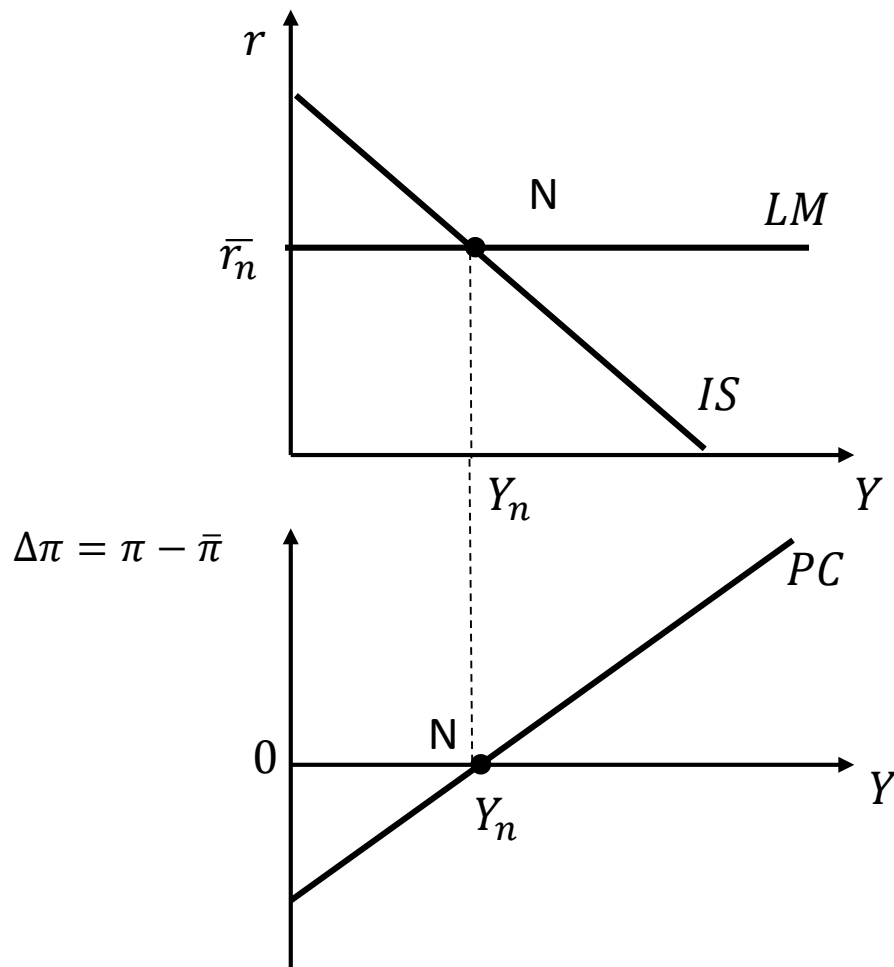
$$Y = Y_n, u = u_n, \Delta\pi^e = 0$$

Expectations are well anchored to the CB target $\pi^e = \bar{\pi}$

$$IS: Y = C(Y - T) + I(Y, r + x) + G$$

$$LM: r = \bar{r}$$

$$PC: \Delta \bar{\pi} = \frac{\alpha}{L} (Y - Y_n)$$



The economy is in a medium-run equilibrium: point N

$$IS \cap LM \text{ at } Y_n$$
$$Y = Y_n, u = u_n$$

Inflation is at the target

$$\Delta \bar{\pi} = 0 \leftrightarrow \pi = \bar{\pi}$$

FEATURES OF THE MEDIUM RUN EQUILIBRIUM

1. THE REAL INTEREST IS AT THE NATURAL LEVEL

$r_n \equiv$ Natural rate of interest (neutral or Wicksellian interest rate)
Interest rate associated with the medium run equilibrium, Y_n
(Goods mkt eq. at the MR)



It is determined implicitly from the equilibrium in the goods market

$$r_n : Y_n = Z_n$$

The goods market is in equilibrium in the medium run if

Aggregate supply = Aggregate demand

$$Y_n = Z_n$$

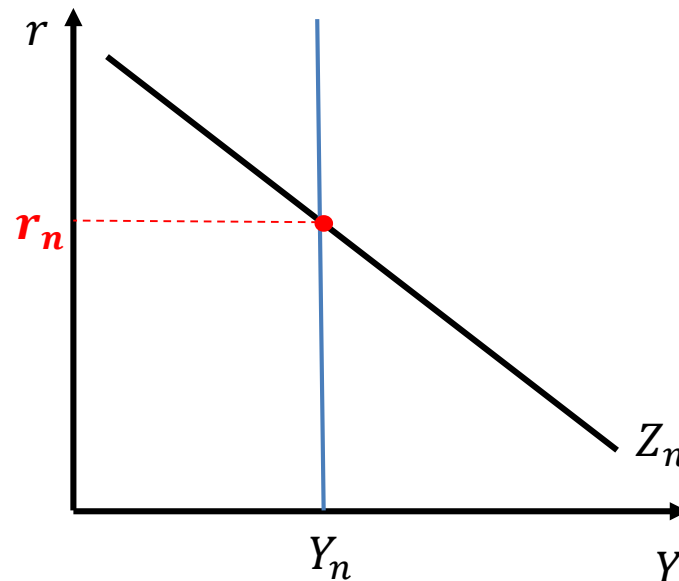
$$Y_n = C(Y_n - T) + \underbrace{I(Y_n, r_n + x) + G}_{Z_n = f(r_n)}$$

$$Y_n = L(1 - u_n) = L \left(1 - \frac{m + z}{\alpha} \right)$$

Determined only by **structural factors** of the:

1. **goods market** m
2. **labor market** z, α
3. **Labor force/population** L

Does not depend on r !!



DETERMINANTS OF r_n

- The natural rate of interest depends on:
 - Y_n
 - G, T : fiscal policy
 - Demand shocks: $c_0, \bar{I}, x, \dots$
- It changes only if the potential output changes and the demand changes permanently
- It does not depend on the monetary policy!!!

2. NOMINAL INTEREST RATE

In the medium run $r = r_n$ and $\pi = \pi^e \Rightarrow$ the nominal interest rate will be
$$i = r_n + \pi^e$$

With anchored expectations $\pi^e = \bar{\pi} \Rightarrow i = r_n + \bar{\pi}$

The higher the inflation target, the higher the nominal interest rate will be

3. INFLATION RATE IS EQUAL TO THE GROWTH RATE OF MONEY

The money market is in equilibrium

$$\frac{M}{P} = L(Y, i)$$

Given that L is constant also the real money supply will be constant

$$g_M = \pi = \bar{\pi}$$

$$\Leftrightarrow$$

$$i = r_n + g_M$$

4. SAVING AND INVESTMENT

In the MR all markets are in equilibrium and expectations are correct

$$Y_n = Z_n \text{ and } I = S_{Nat}$$

$$I(Y_n, r_n + x) = S(Y - T) + T - G$$

5. REAL VARIABLES IN THE MR DO NOT DEPEND ON MONETARY POLICY

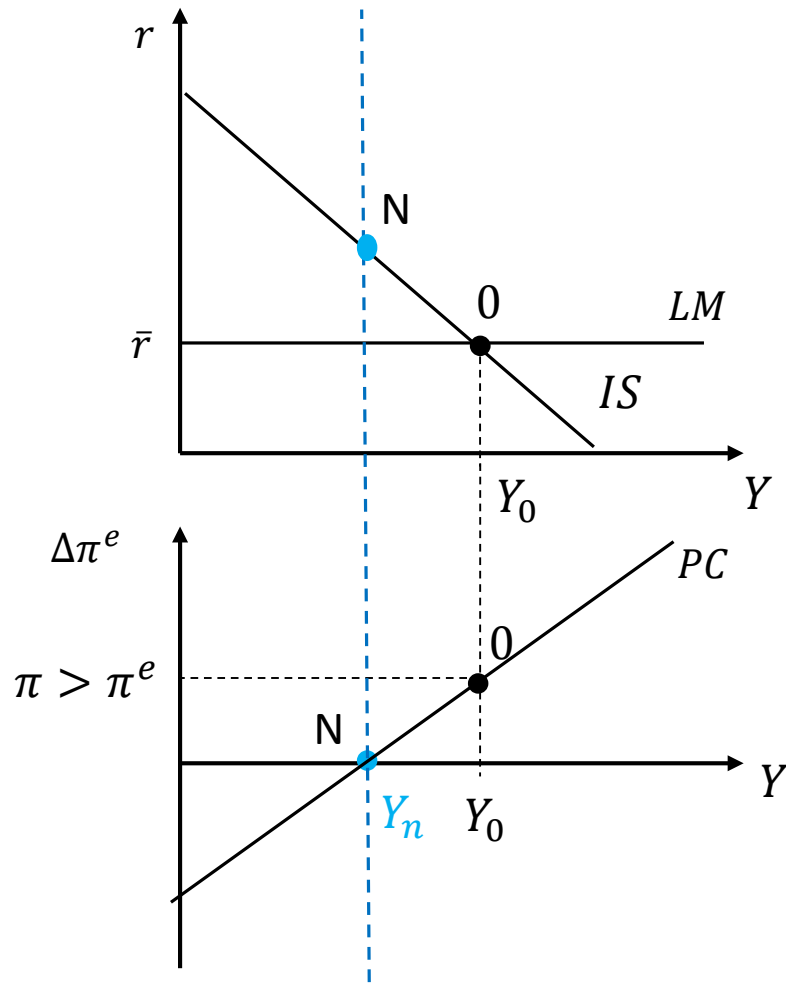
The monetary policy in the medium run determines only the nominal interest rate and inflation

SHORT RUN EQUILIBRIUM

Goods and financial markets are in equilibrium but expectations are not correct => the economy is above/below the with an output gap

$$Y \neq Y_n, u \neq u_n, \Delta\pi^e \neq 0$$

1) ECONOMY ABOVE THE POTENTIAL



$IS \cap LM$ at the right of Y_n

Short Run equilibrium: point 0

$$Y \neq Y_n \Rightarrow \Delta\pi \neq 0$$

Positive output gap: economy above the potential

$$Y > Y_n \Rightarrow u < u_n :$$

$\Rightarrow \uparrow$ workers bargaining power

$\downarrow$ firms bargaining power

$\Rightarrow \uparrow$ nominal W bargained

$\Rightarrow \uparrow$ firms production costs that set higher prices to final consumers

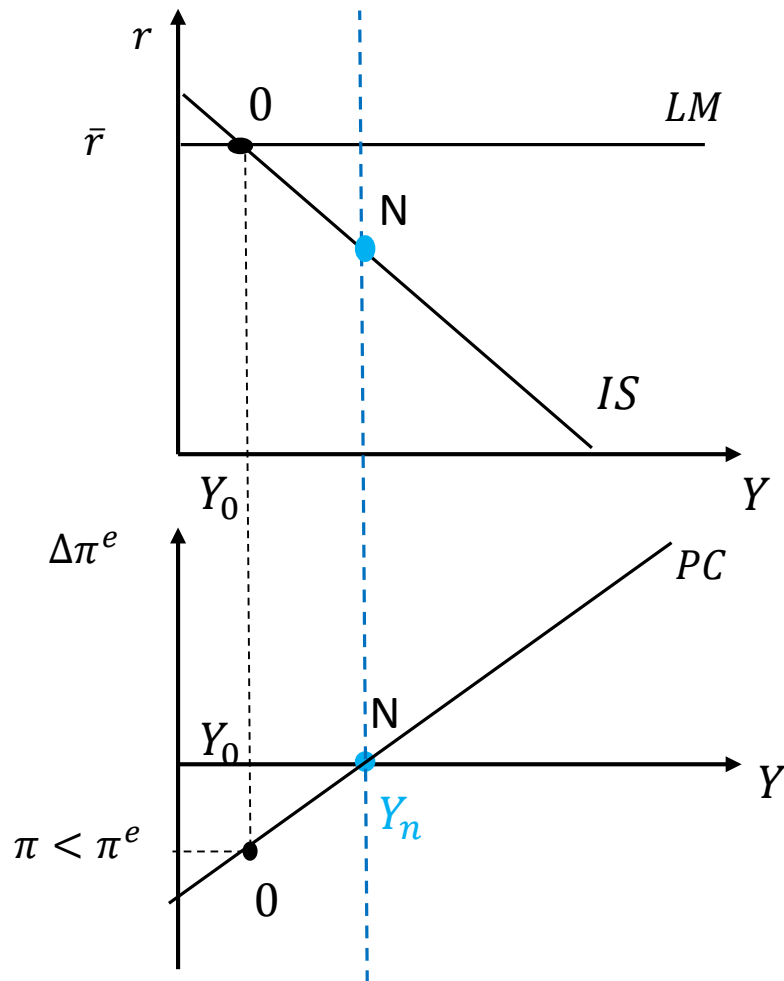
$\Rightarrow \uparrow$ current price level

$\Rightarrow \uparrow$ inflation that will be higher than the CB target expected

$$\Rightarrow \pi > \pi^e$$

$$\pi > \bar{\pi}$$

2) ECONOMY BELOW THE POTENTIAL



$IS \cap LM$ at the left of Y_n

Short Run equilibrium: point 0
 $Y \neq Y_n \Rightarrow \Delta\pi \neq 0$

Negative output gap: economy below the potential

$Y < Y_n \Rightarrow u > u_n$:
 $\Rightarrow \downarrow$ workers bargaining power
 $\uparrow$ firms bargaining power
 $\Rightarrow \downarrow$ nominal W bargained
 $\Rightarrow \downarrow$ firms production costs that set higher prices to final consumers
 $\Rightarrow \downarrow$ current price level
 $\Rightarrow \downarrow$ inflation that will be lower than the CB target expected
 $\Rightarrow \pi < \pi^e$

$$\pi < \bar{\pi}$$

HOW DO WE REACH A MR EQUILIBRIUM?

Nowaday mainly thanks to the interventions of the CB

ECB goal: price stability

FED goal: maximize employment with stable prices

MONETARY POLICY and INFLATION TARGETING

Modern CB have targets in terms of inflation

- **European Central Bank**

- The primary objective of the ECB's monetary policy is to maintain **price stability**
- The Governing Council considers that price stability is best maintained by aiming for **2% inflation over the medium term**.

- **FED:**

- The goals of monetary policy are to promote maximum employment, **stable prices** and moderate long-term interest rates.
- By implementing effective monetary policy, the Fed can maintain **stable prices**, thereby supporting conditions for long-term economic growth and maximum employment

MONETARY POLICY

- Main desirable objectives:
 - A low and stable inflation
 - Production at potential (Y_n)
- Up to the 80s, the strategy of monetary policy was **money targeting**
 - Control the amount of money (growth)
 - Results were unsatisfactory
- Strategy adopted today: ***inflation targeting***
- Two main principles:
 - *Keep inflation stable and low;*
 - Follow, explicitly or implicitly, a **policy rule on the interest rate**

INFLATION TARGETING

- Phillips Curve:

$$\pi - \pi^e = \frac{\alpha}{L}(Y - Y_n)$$

- If the Central Bank is credible in pursuing the aim of an inflation rate equal to $\bar{\pi}$
 - Agents form expectations coherently $\pi^e = \bar{\pi}$
 - The economy is always at the natural rate of unemployment
 - Hence, $Y = Y_n$
- **Divine Coincidence!**
 - A point of reference, but the real world is a bit more complicated ...

MONETARY POLICY IN ACTION

- In practice CB adopt a «flexible» inflation targeting:
 - Uncertainty over the impact of changes in the interest rate on production and inflation
 - A gradual approach => Target over the medium run

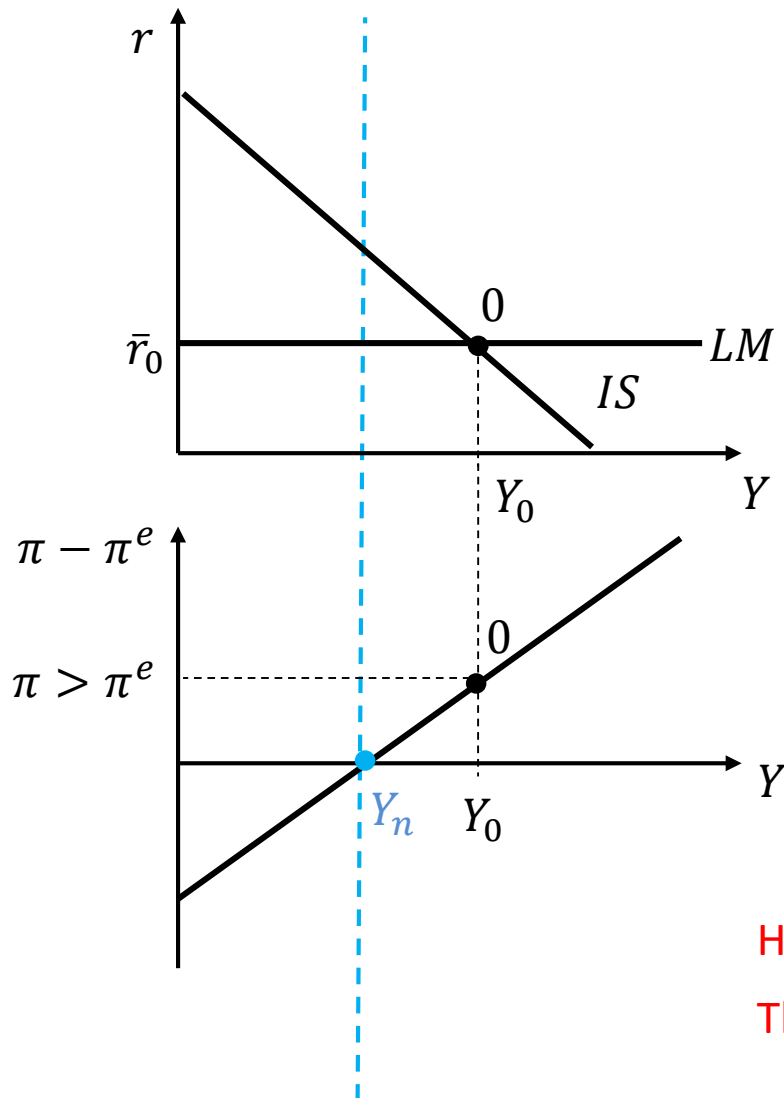
IS-LM-PC: ADJUSTMENT TO THE MR

In 0: short run

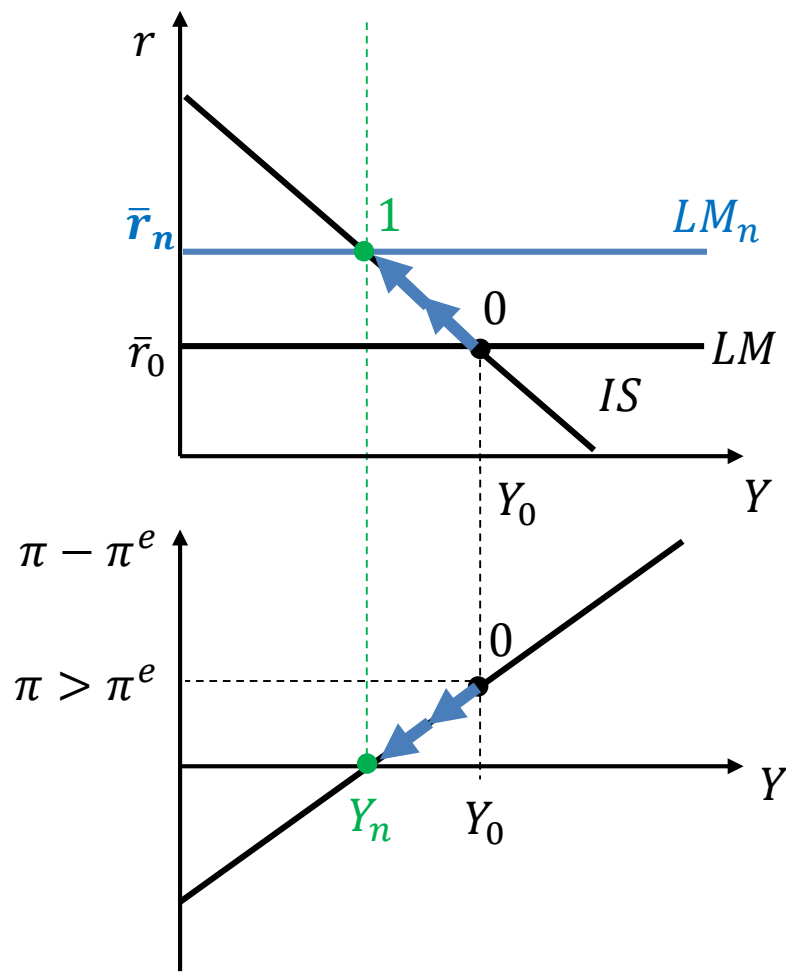
$$Y > Y_n \Rightarrow \pi > \pi^e$$

POSITIVE output gap:
economy ABOVE the
potential

$$W \uparrow \Rightarrow \pi > \pi^e$$



How do we reach the medium run equilibrium?
Through CB interventions to close the output gap



The CB acts to reduce inflation growth

$$i \uparrow \Rightarrow r \uparrow \Rightarrow LM \uparrow$$

$$r \uparrow \Rightarrow I \downarrow \Rightarrow Y \downarrow$$

along the IS

In **1**, the economy is at the
medium run equilibrium:

$$\pi = \pi^e$$

$$Y = Y_n \quad u = u_n$$

$$r = r_n$$

ECONOMIC INTUITION

- CB implements contractionary MP by selling bonds.
 - On the bonds market the price of bond decreases and, given their face value, their nominal return increases.
 - Given expected inflation, also the real interest rate increases
- The higher borrowing rate reduces investment, the aggregate demand and through the multiplier income
- If GDP decreases, employment decreases and unemployment increases reducing the workers bargaining power with respect to firms
- Bargained nominal wages decrease thus reducing firms production costs.
- Firms can maximize their profits by charging lower prices
- The inflation rate decreases

$H \downarrow \Rightarrow \epsilon P_B \downarrow$ and $i \uparrow \Rightarrow$ given expected inflation also $r \uparrow \Rightarrow \downarrow I, \downarrow Z$ and $\downarrow Y \Rightarrow$
 $u \uparrow$ and $W \downarrow \Rightarrow P \downarrow \Rightarrow \pi \downarrow$

In 0: short run

$$Y < Y_n \Rightarrow \pi < \pi^e$$

How do we reach the medium run equilibrium?

The CB acts to increase output

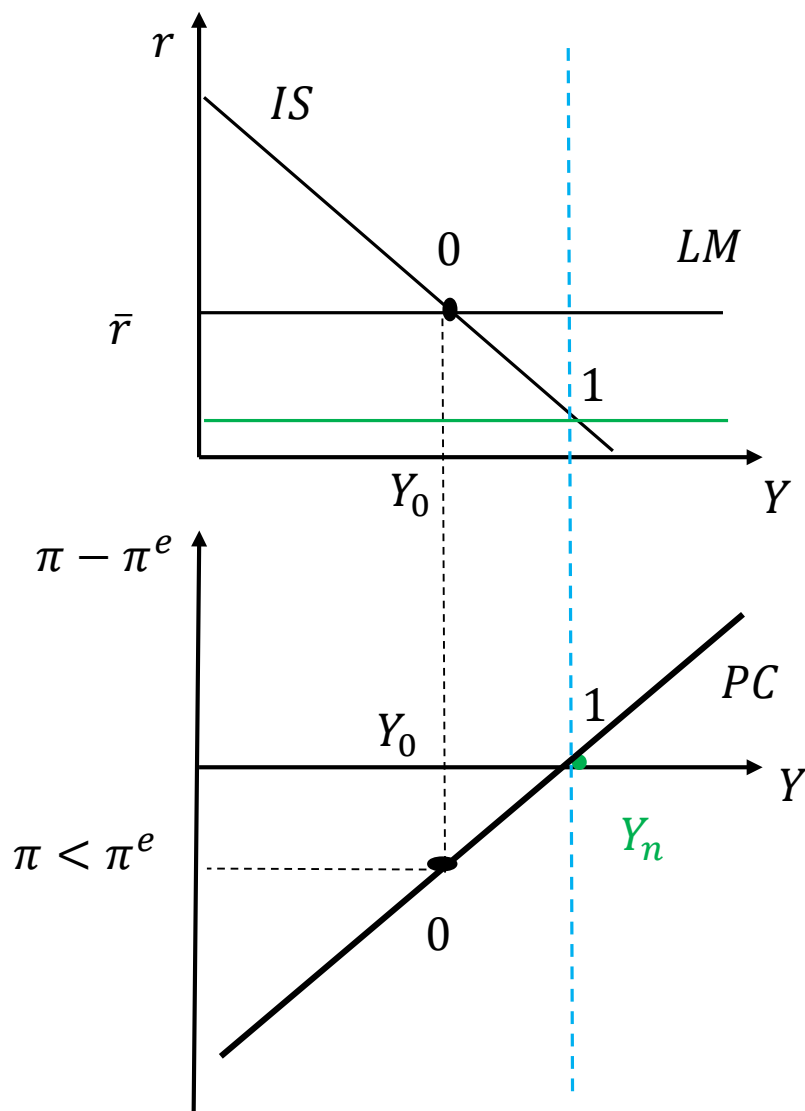
$i \downarrow \Rightarrow r \downarrow \Rightarrow LM \downarrow$
 $r \downarrow \Rightarrow I \uparrow \Rightarrow Y \uparrow$
along the IS

In **1**, the economy is at the
medium run equilibrium:

$$\pi = \pi^e$$

$$Y = Y_n \quad u = u_n$$

$$r = r_n$$



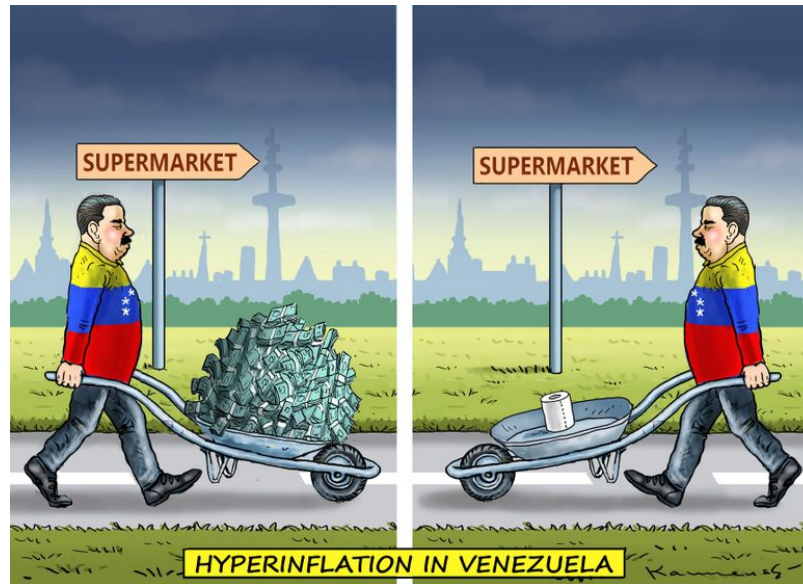
ECONOMIC INTUITION

- CB implements expansionary MP by buying bonds.
 - On the bonds market the price of bond increases and, given their face value, their nominal return decreases.
 - Given expected inflation, also the real interest rate decreases
- The lower borrowing rate increases investment, the aggregate demand and through the multiplier income
- If GDP increases, employment increases and unemployment decreases increasing the workers bargaining power with respect to firms
- Bargained nominal wages increase thus increasing firms production costs.
- Firms in order to maximize their profits charge higher prices
- The inflation rate increases

$H \uparrow \Rightarrow \epsilon P_B \uparrow$ and $i \downarrow \Rightarrow$ given expected inflation also $r \downarrow \Rightarrow \uparrow I, \uparrow Z$ and $\uparrow Y \Rightarrow$
 $u \uparrow$ and $W \downarrow \Rightarrow P \uparrow \Rightarrow \pi \uparrow$

ADJUSTMENT IN THE REAL WORLD

- In the **real world**, the economy goes back to medium run gradually!!!
- Why is it the case?
 1. Not easy to detect whether is different from (not observed!!!)
 2. Difficulties in estimating shocks' magnitude and persistency
 3. Lag in monetary policy's transmission to the real economy



COSTS & BENEFITS OF INFLATION

COSTS OF INFLATION

- **Menu costs** – costs of changing prices due to inflation. e.g. new menus, catalogues
 - Change more frequently
- **Shoe-leather costs** – Costs of behaviors to reduce the impact of inflation
 - Higher opportunity cost of money

COSTS OF INFLATION

- Tax distortions:
 - tax brackets not adjusted in line with inflation
 - over-taxation of capital gains
- Wealth redistribution:
 - E.g. from creditors to borrowers!!!

COSTS OF INFLATION

- **Uncertainty and volatility** – Higher inflation is typically associated with more **volatile** inflation, which increases the risk of financial assets that promise fixed nominal payments in the future
- **Money illusion**: systematic mistakes in comparing nominal vs. real changes
 - If inflation becomes volatile, this comparison becomes even more difficult

BENEFITS OF INFLATION

- Moderate inflation enables adjustment of wages:
 - **Money illusion:** it is difficult to cut nominal wages
 - With positive inflation, it is easier to adjust real wages, without cutting the nominal ones
- Negative (real) interest rates

SUMMARY

- IS-LM-PC Model
 - From short to medium run
 - Role of Central Banks
- Costs and benefits of inflation